Advanced Microeconomics I (Fall 2026)

Homework 1

1. A consumer consumes three goods and satisfies Walras' law. The demand functions are

$$x_1(p,w) = \frac{\alpha p_1}{2(p_1+p_2+p_3)} \cdot \frac{w}{p_1}, \qquad x_2(p,w) = \frac{p_2}{p_1+p_2+p_3} \cdot \frac{w}{p_2}, \qquad x_3(p,w) = \frac{p_3}{p_1+p_2+p_3} \cdot \frac{w}{p_3}.$$

Find the consumer's demand function for good 1 and the value of α. Are the demand functions homogeneous of degree one?

2. A consumer consumes two goods in periods 1 and 2. In period 1, prices are $(p_1, p_2) = (5, 5)$ and the chosen bundle is $(x_1, x_2) = (5, 5)$. In period 2, prices are $(p_1, p_2) = (4, 5)$ and the chosen bundle is $(x_1, x_2) = (x, 6)$.

(1) Over what range of x is the consumption bundle in period 1 revealed preferred to that in period 2? Assume WARP is satisfied.

(2) Over what range of x is the consumption bundle in period 2 revealed preferred to that in period 1? Assume WARP is satisfied.

(3) Over what range of x is the consumer's behavior inconsistent with WARP?

3. The utility function is

$$u(x_1, x_2) = x_1^{1/2} + 2x_2^{1/2}.$$

(a) Find the Marshallian demand functions for goods 1 and 2.

(b) Find the expenditure function and the Hicksian demand functions, and verify Shephard's lemma.

(c) Find the indirect utility function, and verify Roy's identity.
(Hint: You can use Roy's identity to understand the Envelope Theorem independently.)

4. A consumer's demand function for apples is

$$x = 20 + \frac{w}{20p},$$

where x is the quantity demanded, w is income, and p is the price of apples. Initially $(w, p) = (240, 6)$. Suppose the price of apples falls to 4.

(1) Calculate the Slutsky substitution effect of the price change.

(2) Calculate the corresponding income effect.

(Hint: This is different from the Hicksian substitution effect, which keeps utility constant. You can find an example to practice the Hicksian substitution effect independently.)

5. Let $x_i(p, y)$ and $x_j(p, y)$ denote the ordinary demand functions for goods i and j, respectively. Suppose $\eta_i = \eta_j$, that is, the income elasticity of demand for good i is equal to the income elasticity of demand for good j. Show that

$$\frac{\partial x_i(p, y)}{\partial p_j} = \frac{\partial x_j(p, y)}{\partial p_i}.$$